October 30, 2026

$ Research with AI Agents

What Works, What It Costs, and What an Institute Needs

author: Lukas Theissinger

$ Outline

> [1] OpenAI and Agent Usage

> [2] Owning Intelligence and Infrastructure at Lamarr

$ cat outline

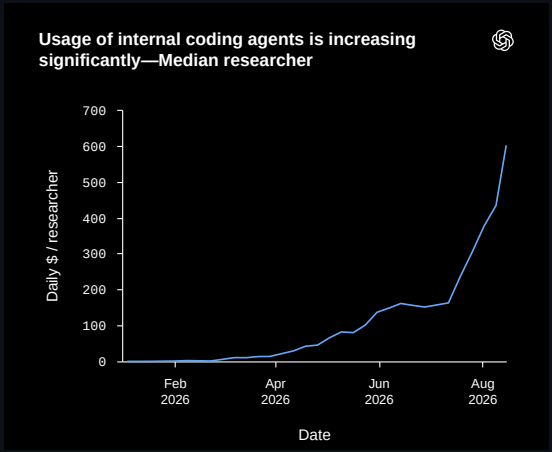
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OpenAI and Agent Usage

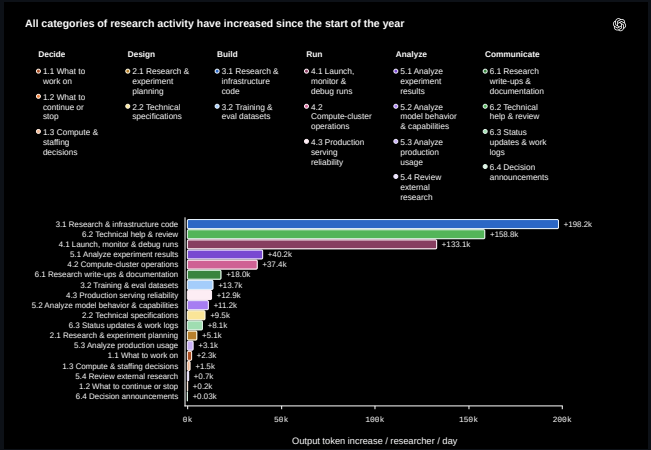
$ FOMO Headline: OpenAI Agent Usage



Median researcher, daily $ spent on internal coding agents. Source: OpenAI, “Research acceleration” (self-reported).

OpenAI and Agent Usage

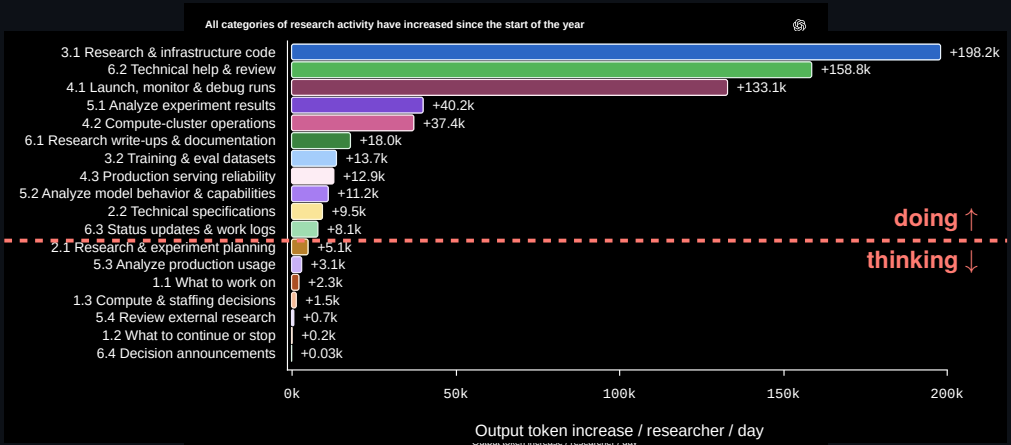
$ What do agents do at OpenAI?



Increase in output tokens per researcher per day, by research activity. Source: OpenAI, “Research acceleration” (self-reported).

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OpenAI and Agent Usage

$ Dettmers' Interpretation

What the OpenAI chart suggests

All 18 activities went up: agents are used everywhere,
but how much and how exactly is still open.

Sources: timdettmers.com/2026/01/13/use-agents-or-be-left-behind; OpenAI, "Research acceleration" (self-reported).

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Frequent, repetitive work grew most; rare decisions grew least.	Automate only if how often $\times$ time saved $>$ time to build.
Did agents take over the *thinking*? We can't say: a plan or a decision is short, so tokens are the wrong measure.	Agent use is a *skill*: not everything can or should be automated (degree of automation).

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OpenAI and Agent Usage

$ Field Report: Automating Exercises

1. **The problem.** PhD student, tutoring an exercise class every two weeks:
   - 2 h exercise class + 6 h preparation (solving the tasks, making the slides)
   - rough estimate: **2 h + 6 h = 8 h every two weeks**

OpenAI and Agent Usage

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4. **Ask your agent to try it:** `github.com/Newida/AutomatingExercises` **Avoid:** Don't do anything BY HAND on this project. If you want to change it, ASK YOUR AGENT to change it.

$ cat outline

[1] OpenAI and Agent Usage

> [2] Owning Intelligence and Infrastructure at Lamarr

$

Owning Intelligence and Infrastructure at Lamarr

$ Automating at Bigger Scale

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Example: travel expense reports

Collect receipts, fill in the university form, look up per-diem rules, fix what admin sends back: say **3 trips** $\times$ **3 h per person per year**. An agent that reads the receipts and files the form correctly needs $\sim$**200 h** (rules, forms, access to the admin system).

$$\frac{\text{how often} \times \text{time saved}}{\text{time to build}}$$

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institute, 10 000 people	$10\,000 \times 9$ h = 90 000 h/year	200 h	**pays $>400\times$**

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Dear management: do not assume that employees will continue spending their own money. We are bottlenecking ourselves: we don't want to spend 100 € per month just to do our work, so we spend 20 €, effectively leaving performance on the table. Also, people will just switch to other institutes, even outside Germany, that supply them with the AI they want and need, for free, to get important work done.

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So let's talk about costs.

Owning Intelligence and Infrastructure at Lamarr

$ Decision 1: Subscription or API?

Subscription (20–200 €/month)

- \+ flat price, no surprise bills
- \+ very cheap for heavy interactive use
- \- usage caps (5-hour and weekly)
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Our comparison: one month of my usage, priced at API list prices

	subscription	same usage via API	factor
Claude	20 $ (Pro)	∼680 $	∼34×
Codex	20 $ (Plus)	∼145 $	∼7×

But one batch job, 27 benchmark tasks via API, cost only **2.25 $**.

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Rule of thumb: people → subscription; pipelines and batch jobs → API.

Owning Intelligence and Infrastructure at Lamarr

$ Decision 2: Self-Hosting Open Models

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Small open models are clearly behind. The largest open models are much closer to the frontier; on our tasks they are **still untested**.

Owning Intelligence and Infrastructure at Lamarr

$ Self-Hosting for the Institute: Pros and Cons

Quality and performance

+ data never leaves Lamarr (privacy, unpublished work, GDPR)
+ no vendor caps or outages, models don't change under you
+ fine-tune on our own data, research on the models themselves

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Owning Intelligence and Infrastructure at Lamarr

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+ Lamarr already has the GPUs and the people who run them

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Rough comparison

2 nodes × 300 k€ over 4 years ≈ 12.5 k€/month ≈ **300 people** × **40 €** ≈ **125 people** × **100 €**

With Lamarr's compute, a frontier-class open model for everyone is realistic: it pays off through privacy, scale and no caps; the closed models keep a lead of months.

Owning Intelligence and Infrastructure at Lamarr

$ Takeaways

For the individual

1. **You need to use AI agents.**

Owning Intelligence and Infrastructure at Lamarr

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1. You need to use AI agents.
2. Use AI agents to change your bottlenecks at work

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